

Paul He

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Education

University of Toronto, BSc in Computer Science (Honours) Sep 2021 – May 2025
ETH Zürich, Exchange Year in Computer Science Sep 2023 – Aug 2024

Research Interests

Natural Language Processing (NLP), Large Language Models (LLMs), Explainable Artificial Intelligence, Parsing

Research Experience

Overcoming Theoretical Limitations of Large Language Models Jan 2025 - Present

Supervisor: Prof. Dr. Gerald Penn

- Building on existing methods to overcome theoretical limitations of self-attention and exploring ways to efficiently improve language model performance.

Causal Large Language Models Jan 2025 - Present

Supervisor: Prof. Dr. Zhijing Jin

- Exploring various methods and evaluation metrics to improve Large Language Model's ability to formalize Causal questions.
- Improving multilingual language model interpretability.

Language Model Interpretability Jun 2024 – Sep 2024

Supervisors: Tianyu Liu, Prof. Dr. Ryan Cotterell

- Investigated attribution methods such as the change in Jensen-Shannon divergence for Fisher information as an approximation to attention scores to identify the part of context the language model focuses on.
- Developed and implement a mechanism to increase the accuracy of open language models by up to 3% in question-answering tasks on NQ-Open and ELI5 datasets.

Publications

Likelihood as a Performance Gauge for Retrieval-Augmented Generation 2024

Tianyu Liu, Jirui Qi, **Paul He**, Arianna Bisazza, Mrinmaya Sachan, Ryan Cotterell

NAACL 2025 (Main), <https://arxiv.org/abs/2411.07773>

Projects

Parsing Combinatory Categorical Grammars (CCGs) Sep 2024 – Dec 2024

Supervisor: Prof. Dr. Gerald Penn

- Implemented the worlds first public CCG parser introduced by Kuhlmann and Satta.

Speeding up Earley's Algorithm Mar 2024 - Aug 2024

- Proposed and implemented a grid search approach, which resulted in an algorithm $\times 5$ faster than the naive algorithm.
- Tested and experimented with the proposed algorithms on Treebank datasets to validate theoretical findings.

Word Segmentation and Part-of-Speech Tagging with Transformer Oct 2023 - Jan 2024

- Designed a new decoder embedding mechanism to better capture subtle linguistic nuances and dependencies.
- Implemented the mechanism and achieved state-of-the-art results with a validation accuracy of 96.29%.

Skills

Programming Languages: Python, C, Java, SQL

Spoken Languages: English (Fluent), Mandarin (Proficient), German (Proficient)